Homework Solutions: Lectures 16–18

Laplace Transform: Definition, Inversion, Properties, and System Analysis

CEC 315 – Signals and Systems

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Problem 1: Computing Laplace Transforms and ROCs

Part (a): $x_1(t) = 4e^{-2t}\,u(t)$

Step 1: Apply the standard pair $e^{-at}\,u(t)\mathcal{L}\frac{1}{s+a}$, ROC: $\mathrm{Re}\{s\} > -a$.

Here $a = 2$, so:

$$\boxed{X_1(s) = \frac{4}{s+2}\,, \qquad \mathrm{Re}\{s\} > -2}$$

Part (b): $x_2(t) = -3e^{5t}\,u(-t)$

Step 1: Rewrite as $-3 \cdot \big[-e^{-(-5)t}\,u(-t)\big] \cdot (-1)$. Using the pair $-e^{-at}\,u(-t)\mathcal{L}\frac{1}{s+a}$ with ROC $\mathrm{Re}\{s\} < -a$:

Here $a = -5$:

$$-e^{5t}\,u(-t) \;\; \mathcal{L} \;\; \frac{1}{s-5}\,, \quad \mathrm{Re}\{s\} < 5$$

Multiply by 3:

$$\boxed{X_2(s) = \frac{3}{s-5}\,, \qquad \mathrm{Re}\{s\} < 5}$$

Part (c): $x_3(t) = 7\delta(t-3)$

Step 1: Use the pair $\delta(t-t_0)\mathcal{L}e^{-st_0}$, ROC: all s.

$$\boxed{X_3(s) = 7e^{-3s}\,, \qquad \text{ROC: all } s}$$

Part (d): $x_4(t) = te^{-4t}\,u(t)$

Step 1: Use the pair $t\,e^{-at}\,u(t)\mathcal{L}\frac{1}{(s+a)^2}$, ROC: $\mathrm{Re}\{s\} > -a$.

With $a = 4$:

$$\boxed{X_4(s) = \frac{1}{(s+4)^2}\,, \qquad \mathrm{Re}\{s\} > -4}$$

Part (e): $x_5(t) = 2e^{-t}\,u(t) - 5e^{-3t}\,u(t)$

Step 1: Transform each term:

$$2e^{-t}\,u(t) \;\mathcal{L}\; \frac{2}{s+1}\,, \quad \mathrm{Re}\{s\} > -1$$
$$-5e^{-3t}\,u(t) \;\mathcal{L}\; \frac{-5}{s+3}\,, \quad \mathrm{Re}\{s\} > -3$$

Step 2: Add and find common denominator:

$$X_5(s) = \frac{2}{s+1} - \frac{5}{s+3} = \frac{2(s+3) - 5(s+1)}{(s+1)(s+3)} = \frac{2s+6-5s-5}{(s+1)(s+3)} = \frac{-3s+1}{(s+1)(s+3)}$$

Step 3: ROC is the intersection $\mathrm{Re}\{s\} > -1 \;\cap\; \mathrm{Re}\{s\} > -3 = \mathrm{Re}\{s\} > -1$.

$$\boxed{X_5(s) = \frac{-3s+1}{(s+1)(s+3)}\,, \qquad \mathrm{Re}\{s\} > -1}$$

Poles: $s = -1$ and $s = -3$. Zero: $s = 1/3$.

Part (f): $x_6(t) = e^{-2t}\,u(t) + 3e^{4t}\,u(-t)$

Step 1: Transform each term:

$$e^{-2t}\,u(t) \;\mathcal{L}\; \frac{1}{s+2}\,, \quad \mathrm{Re}\{s\} > -2$$
$$3e^{4t}\,u(-t) = -3 \cdot \big[-e^{4t}\,u(-t)\big] \;\mathcal{L}\; \frac{-3}{s-4}\,, \quad \mathrm{Re}\{s\} < 4$$

Step 2: Add and find the ROC:

$$X_6(s) = \frac{1}{s+2} - \frac{3}{s-4} = \frac{(s-4) - 3(s+2)}{(s+2)(s-4)} = \frac{-2s-10}{(s+2)(s-4)}$$

Step 3: ROC is the intersection $\sigma > -2 \;\cap\; \sigma < 4$:

$$\boxed{X_6(s) = \frac{-2s-10}{(s+2)(s-4)} = \frac{-2(s+5)}{(s+2)(s-4)}\,, \qquad -2 < \mathrm{Re}\{s\} < 4}$$

Step 4: The $j\omega$-axis ($\sigma = 0$) lies in the strip $-2 < \sigma < 4$, so the Fourier transform of $x_6(t)$ **does exist**.

Key Point

The Fourier transform exists whenever the $j\omega$-axis is contained in the ROC. For this two-sided signal the ROC is a strip, and $\sigma = 0$ falls inside it.

Problem 2: Inverse Laplace Transform via Partial Fractions

Part (a): Distinct Real Poles, Right-Sided

Given: $X(s) = \dfrac{3s+5}{(s+1)(s+4)}$, ROC: $\mathrm{Re}\{s\} > -1$.

Step 1: Partial fractions.

$$\frac{3s+5}{(s+1)(s+4)} = \frac{A}{s+1} + \frac{B}{s+4}$$

Cover-up:

- $s = -1$: $3(-1)+5 = A(-1+4) \;\Rightarrow\; 2 = 3A \;\Rightarrow\; A = 2/3$.
- $s = -4$: $3(-4)+5 = B(-4+1) \;\Rightarrow\; -7 = -3B \;\Rightarrow\; B = 7/3$.

Step 2: Determine directions. ROC is $\sigma > -1$. Both poles ($s = -1$ and $s = -4$) are to the left of the ROC boundary, so both terms are right-sided.

Step 3: Invert.

$$\boxed{x(t) = \frac{2}{3}\,e^{-t}\,u(t) + \frac{7}{3}\,e^{-4t}\,u(t)}$$

Check: $x(0^+) = 2/3 + 7/3 = 3$. Initial value theorem: $\lim_{s\to\infty} sX(s) = \lim_{s\to\infty} \frac{3s^2+5s}{s^2+5s+4} = 3$. ✓

Part (b): Distinct Real Poles, Two-Sided

Given: $X(s) = \dfrac{2s-4}{(s+2)(s-3)}$, ROC: $-2 < \mathrm{Re}\{s\} < 3$.

Step 1: Partial fractions.

$$\frac{2s-4}{(s+2)(s-3)} = \frac{A}{s+2} + \frac{B}{s-3}$$

- $s = -2$: $2(-2)-4 = A(-2-3) \;\Rightarrow\; -8 = -5A \;\Rightarrow\; A = 8/5$.
- $s = 3$: $2(3)-4 = B(3+2) \;\Rightarrow\; 2 = 5B \;\Rightarrow\; B = 2/5$.

Step 2: Determine directions from the ROC strip $-2 < \sigma < 3$:

- Pole at $s = -2$: the ROC is to the **right** of this pole $\Rightarrow$ right-sided term.
- Pole at $s = 3$: the ROC is to the **left** of this pole $\Rightarrow$ left-sided term.

Step 3: Invert.

$$\frac{8/5}{s+2}\ \text{(right-sided)} \;\longrightarrow\; \frac{8}{5}\,e^{-2t}\,u(t)$$

$$\frac{2/5}{s-3}\ \text{(left-sided)} \;\longrightarrow\; -\frac{2}{5}\,e^{3t}\,u(-t)$$

$$\boxed{x(t) = \frac{8}{5}\,e^{-2t}\,u(t) - \frac{2}{5}\,e^{3t}\,u(-t)}$$

Key Point

The left-sided term uses the pair $\frac{1}{s-a}$ with ROC $\sigma < a \;\longleftrightarrow\; -e^{at}\,u(-t)$. That negative sign in front is part of the pair, not an error.

Part (c): Repeated Poles

Given: $X(s) = \dfrac{6}{(s+2)^2(s+5)}$, ROC: $\mathrm{Re}\{s\} > -2$.

Step 1: Partial fractions.

$$\frac{6}{(s+2)^2(s+5)} = \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{s+5}$$

Multiply through by $(s+2)^2(s+5)$:

$$6 = A(s+2)(s+5) + B(s+5) + C(s+2)^2$$

Find C: set $s = -5$: $6 = C(-5+2)^2 = 9C \;\Rightarrow\; C = 2/3$.

Find B: set $s = -2$: $6 = B(-2+5) = 3B \;\Rightarrow\; B = 2$.

Find A: compare s^2 coefficients: $0 = A + C \;\Rightarrow\; A = -C = -2/3$.

Step 2: All poles at $\sigma \le -2$, ROC to the right, so all terms are right-sided.

Step 3: Invert using $\frac{1}{(s+a)^2} \leftrightarrow t\,e^{-at}\,u(t)$:

$$\boxed{x(t) = \left[-\frac{2}{3}\,e^{-2t} + 2\,t\,e^{-2t} + \frac{2}{3}\,e^{-5t}\right] u(t)}$$

Check: $x(0^+) = -2/3 + 0 + 2/3 = 0$. Initial value theorem: $\lim_{s\to\infty} sX(s) = \lim_{s\to\infty} \frac{6s}{(s+2)^2(s+5)} = 0$. ✓

Part (d): Complex Conjugate Poles

Given: $X(s) = \dfrac{s+3}{s^2+6s+25}$, ROC: $\mathrm{Re}\{s\} > -3$.

Step 1: Complete the square.

$$s^2 + 6s + 25 = (s+3)^2 + 16 = (s+3)^2 + 4^2$$

Poles at $s = -3 \pm j4$. Damping rate: $\sigma_0 = 3$. Oscillation frequency: $\omega_d = 4$ rad/s.

Step 2: Rewrite the numerator.

$$s + 3 = (s + 3) + 0$$

So:

$$X(s) = \frac{(s+3)}{(s+3)^2 + 4^2}$$

Step 3: Invert using $\dfrac{s+a}{(s+a)^2 + \omega_d^2} \leftrightarrow e^{-at}\cos(\omega_d t)\,u(t)$:

$$\boxed{x(t) = e^{-3t}\cos(4t)\,u(t)}$$

This is a damped sinusoid decaying with time constant $\tau = 1/3$ s, oscillating at 4 rad/s.

Problem 3: Laplace Transform Properties

Part (a): s-Domain Shift

Given: $\cos(5t)\,u(t)\mathcal{L}\dfrac{s}{s^2+25}$, ROC: $\sigma > 0$.

The s-domain shift property says: multiplying by e^{-2t} in time replaces s by $s+2$ in the transform.

$$\boxed{F(s) = \frac{s+2}{(s+2)^2+25}\,, \qquad \mathrm{Re}\{s\} > -2}$$

The ROC shifts from $\sigma > 0$ to $\sigma > 0 - 2 = -2$.

Part (b): Differentiation in Time

Given: $X(s) = \dfrac{1}{s+3}$, ROC: $\sigma > -3$. So $x(t) = e^{-3t}\,u(t)$.

Step 1: By the differentiation property:

$$Y(s) = s\,X(s) = \frac{s}{s+3} = 1 - \frac{3}{s+3}$$

(where the last step uses polynomial division: $\frac{s}{s+3} = \frac{(s+3)-3}{s+3}$).

ROC: $\mathrm{Re}\{s\} > -3$ (at least).

Step 2: Verify by computing $y(t) = dx/dt$ explicitly.

$x(t) = e^{-3t}\,u(t)$, so:

$$y(t) = \frac{d}{dt}\left[e^{-3t}\,u(t)\right] = -3e^{-3t}\,u(t) + e^{-3\cdot 0}\,\delta(t) = -3e^{-3t}\,u(t) + \delta(t)$$

Taking the Laplace transform of this:

$$Y(s) = \frac{-3}{s+3} + 1 = \frac{-3+(s+3)}{s+3} = \frac{s}{s+3} \quad \checkmark$$

Part (c): Convolution Property

Step (i): $H_1(s) = \dfrac{1}{s+1}$, ROC: $\sigma > -1$. $\quad H_2(s) = \dfrac{3}{s+6}$, ROC: $\sigma > -6$.

Step (ii):

$$H(s) = H_1(s)\cdot H_2(s) = \frac{3}{(s+1)(s+6)}\,, \qquad \mathrm{Re}\{s\} > -1$$

Step (iii): Partial fractions.

$$\frac{3}{(s+1)(s+6)} = \frac{A}{s+1} + \frac{B}{s+6}$$

- $s = -1$: $3 = A(5) \Rightarrow A = 3/5$.
- $s = -6$: $3 = B(-5) \Rightarrow B = -3/5$.

$$\boxed{h(t) = \frac{3}{5}\, e^{-t}\, u(t) - \frac{3}{5}\, e^{-6t}\, u(t) = \frac{3}{5}\left(e^{-t} - e^{-6t}\right) u(t)}$$

Check: $h(0^+) = 3/5 - 3/5 = 0$ ✓ (convolution of two causal signals starts at zero).

Part (d): Initial and Final Value Theorems

Given: $X(s) = \dfrac{8(s+1)}{s(s+2)(s+4)}$, ROC: $\sigma > 0$.

(i) Initial value:

$$x(0^+) = \lim_{s\to\infty} s\,X(s) = \lim_{s\to\infty} \frac{8s(s+1)}{s(s+2)(s+4)} = \lim_{s\to\infty} \frac{8s^2+8s}{s^3+6s^2+8s} = 0$$

$$\boxed{x(0^+) = 0}$$

(ii) Final value check: $sX(s) = \dfrac{8(s+1)}{(s+2)(s+4)}$ has poles at $s = -2$ and $s = -4$, both in the LHP. The theorem applies. ✓

$$x(\infty) = \lim_{s\to 0} \frac{8(s+1)}{(s+2)(s+4)} = \frac{8(1)}{(2)(4)} = 1$$

$$\boxed{x(\infty) = 1}$$

(iii) Full inversion.

$$\frac{8(s+1)}{s(s+2)(s+4)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+4}$$

- $s = 0$: $8(1) = A(2)(4) \Rightarrow A = 1$.
- $s = -2$: $8(-1) = B(-2)(2) \Rightarrow -8 = -4B \Rightarrow B = 2$.
- $s = -4$: $8(-3) = C(-4)(-2) \Rightarrow -24 = 8C \Rightarrow C = -3$.

$$\boxed{x(t) = \left[1 + 2e^{-2t} - 3e^{-4t}\right] u(t)}$$

Verify: $x(0^+) = 1 + 2 - 3 = 0$ ✓. $\quad x(\infty) = 1 + 0 - 0 = 1$ ✓.

Problem 4: System Analysis with $H(s)$

System: $\dfrac{d^2y}{dt^2} + 6\dfrac{dy}{dt} + 8y = 2x(t)$.

Part (a): System Function

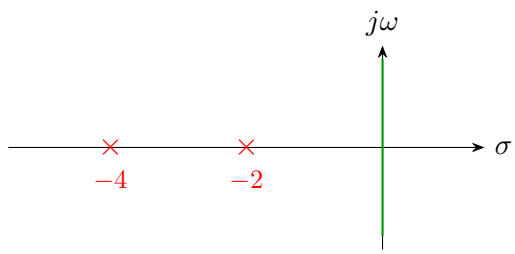
Replace $d^k/dt^k \to s^k$:

$$s^2Y + 6sY + 8Y = 2X \quad \Rightarrow \quad H(s) = \frac{2}{s^2+6s+8} = \frac{2}{(s+2)(s+4)}$$

Poles: $s = -2$ and $s = -4$. No finite zeros. Gain constant: $K = 2$.

$$\boxed{H(s) = \frac{2}{(s+2)(s+4)}, \qquad \mathrm{Re}\{s\} > -2 \text{ (causal)}}$$

Part (b): Pole-Zero Plot and Stability



Both poles are in the open LHP ($\mathrm{Re}\{p_i\} < 0$). Since the system is causal, it is **BIBO stable**. ✓

Part (c): Impulse Response

$$\frac{2}{(s+2)(s+4)} = \frac{A}{s+2} + \frac{B}{s+4}$$

- $s = -2$: $2 = A(2) \Rightarrow A = 1$.
- $s = -4$: $2 = B(-2) \Rightarrow B = -1$.

$$\boxed{h(t) = \left(e^{-2t} - e^{-4t}\right) u(t)}$$

Part (d): Output for $x(t) = e^{-t}\,u(t)$

(i) $X(s) = \dfrac{1}{s+1}$, ROC: $\sigma > -1$.

$$Y(s) = H(s) \cdot X(s) = \frac{2}{(s+1)(s+2)(s+4)}, \quad \mathrm{Re}\{s\} > -1$$

(ii) Partial fractions.

$$\frac{2}{(s+1)(s+2)(s+4)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+4}$$

- $s = -1$: $2 = A(1)(3) \Rightarrow A = 2/3$.
- $s = -2$: $2 = B(-1)(2) \Rightarrow B = -1$.
- $s = -4$: $2 = C(-3)(-2) \Rightarrow C = 1/3$.

$$\boxed{y(t) = \left[\frac{2}{3}\,e^{-t} - e^{-2t} + \frac{1}{3}\,e^{-4t}\right] u(t)}$$

(iii) Verify:

$y(0^+) = 2/3 - 1 + 1/3 = 0$ ✓ (initially at rest).

$y(\infty) = 0$ ✓ (all exponentials decay).

Final-value theorem: $\lim_{s\to 0} sY(s) = \lim_{s\to 0} \frac{2s}{(s+1)(s+2)(s+4)} = 0$ ✓.

Part (e): Stability of $G(s) = 3/(s-1)$

The pole is at $s = 1$ (in the RHP).

If causal: ROC is $\mathrm{Re}\{s\} > 1$. The $j\omega$-axis ($\sigma = 0$) is **not** in the ROC, so the system is **unstable**. The impulse response $g(t) = 3e^t\,u(t)$ grows without bound.

If anti-causal (left-sided): ROC is $\mathrm{Re}\{s\} < 1$. The $j\omega$-axis ($\sigma = 0$) **is** in the ROC, so the system is **BIBO stable**. The impulse response $g(t) = -3e^t\,u(-t)$ decays for $t < 0$.

Key Point

Stability depends on the ROC, not just the pole location. A pole in the RHP makes a *causal* system unstable, but the same pole is compatible with a stable *anti-causal* system. This is why specifying causality (or the ROC) matters.

Problem 5: The Unilateral Laplace Transform

Part (a): First-Order IVP

System: $y' + 4y = 3e^{-t}\,u(t), \quad y(0^-) = 2.$

(i) Transform both sides.

Left side: $\mathcal{L}_u\{y'\} + 4\mathcal{L}_u\{y\} = [sY(s) - y(0^-)] + 4Y(s) = sY - 2 + 4Y.$

Right side: $\mathcal{L}_u\{3e^{-t}\,u(t)\} = \dfrac{3}{s+1}.$

So:

$$(s+4)Y(s) - 2 = \frac{3}{s+1}$$

(ii) Solve for $Y(s)$:

$$Y(s) = \frac{3}{(s+1)(s+4)} + \frac{2}{s+4} = \frac{3 + 2(s+1)}{(s+1)(s+4)} = \frac{2s+5}{(s+1)(s+4)}$$

(iii) Partial fractions.

$$\frac{2s+5}{(s+1)(s+4)} = \frac{A}{s+1} + \frac{B}{s+4}$$

- $s = -1$: $2(-1) + 5 = A(3) \Rightarrow A = 1.$
- $s = -4$: $2(-4) + 5 = B(-3) \Rightarrow B = 1.$

$$\boxed{y(t) = \left(e^{-t} + e^{-4t}\right) u(t)}$$

(iv) Verify:

$y(0) = 1 + 1 = 2 = y(0^-)$ ✓.

$y(\infty) = 0 + 0 = 0$ ✓ (both exponentials decay).

Extra check: $y'(0^+) = -1 - 4 = -5$. From the ODE: $y'(0^+) = -4y(0) + 3e^0 = -8 + 3 = -5$ ✓.

Part (b): Second-Order IVP

System: $y'' + 3y' + 2y = 0, \quad y(0^-) = 1, \quad y'(0^-) = -1.$

(i) Transform:

$$\begin{aligned}
\mathcal{L}_u\{y''\} &= s^2Y - sy(0^-) - y'(0^-) = s^2Y - s + 1 \\
\mathcal{L}_u\{3y'\} &= 3[sY - y(0^-)] = 3sY - 3 \\
\mathcal{L}_u\{2y\} &= 2Y
\end{aligned}$$

Combine:

$$s^2Y - s + 1 + 3sY - 3 + 2Y = 0$$

$$(s^2 + 3s + 2)Y = s + 2$$
$$Y(s) = \frac{s+2}{s^2+3s+2} = \frac{s+2}{(s+1)(s+2)} = \frac{1}{s+1}$$

(ii) Invert:

$$\boxed{y(t) = e^{-t}\,u(t)}$$

Verify: $y(0) = 1$ ✓. $y'(t) = -e^{-t}\,u(t) + \delta(t)$, so $y'(0^-) = -1$ ✓.

(iii) The input is zero ($x(t) = 0$), so there is no zero-state response. This is **purely a zero-input response**: the output is entirely due to the initial conditions.

Key Point

The cancellation of the $(s+2)$ factor between numerator and denominator is not a coincidence. It happens because the initial conditions were chosen so that only one of the two natural modes (e^{-t} and e^{-2t}) is excited. The IC $y'(0^-) = -1 = -y(0^-)$ exactly suppresses the e^{-2t} mode.

Part (c): ZSR/ZIR Decomposition for Part (a)

From part (a): $Y(s) = \dfrac{2s+5}{(s+1)(s+4)}$.

Zero-state response (set $y(0^-) = 0$, keep input):

$$Y_{\text{ZS}}(s) = \frac{3}{(s+1)(s+4)}$$

Partial fractions:

- $s = -1$: $3 = A(3) \Rightarrow A = 1$.
- $s = -4$: $3 = B(-3) \Rightarrow B = -1$.

$$\boxed{y_{\text{ZS}}(t) = \left(e^{-t} - e^{-4t}\right) u(t)}$$

Zero-input response (set input to 0, keep $y(0^-) = 2$):

$$(s+4)Y_{\text{ZI}} = 2 \quad \Rightarrow \quad Y_{\text{ZI}}(s) = \frac{2}{s+4}$$

$$\boxed{y_{\text{ZI}}(t) = 2e^{-4t}\,u(t)}$$

Verify the sum:

$$y_{\text{ZS}}(t) + y_{\text{ZI}}(t) = \left(e^{-t} - e^{-4t}\right) u(t) + 2e^{-4t}\,u(t) = \left(e^{-t} + e^{-4t}\right) u(t) = y(t) \quad ✓$$

Key Point

The ZSR starts at zero ($y_{\text{ZS}}(0) = 0$) because the system is initially at rest. The ZIR carries the initial condition ($y_{\text{ZI}}(0) = 2$). Their sum gives the full response.